



No Code Visualization Tools Power BI

DSCI 454
Week 3, Lecture 1

What are we learning this lecture?



- Exploratory vs Explanatory Visualization
 - Visualization as a thinking tool vs communication tool.
 - Examples of both approaches.
 - Implementation in PBI
 - Filtering
 - Slicing
 - Highlighting
- No Code vs Low Code vs Pro Code Visualization
 - More Week 6 onward
- In-Class Demo / Student Follow Along



Exploratory vs Explanatory Visualization

Exploratory Visualization



- **Reporting**, not storytelling; “All 100 **oysters**”, not pearls
- Trying to **discover** 
 - Trends
 - Relationships
 - Causations vs Correlations
 - Anomalies / Outliers
- Might make many different chart types and not publish or present any of them
- Plot multiple combinations of features/columns vs one another
 - Compare and find relationships
 - Feature selection
- For internal analyst team only; Less polish permissible
- Data, evidence, for data-driven decisions, back-up slides



Explanatory Visualization

- **Storytelling**, not reporting; **Pearls**, not oysters
- Trying to **highlight**
 - Trends
 - Relationships
 - Causations vs Correlations
 - Anomalies / Outliers
 - Found during your exploratory analysis / visualization
- Out of the 100+ exploratory charts, might choose **one or two** to present
- Highlight relationships / correlations to support Ask
- For presentation to externals / customers / executives; Polish

Hybrid Dashboards



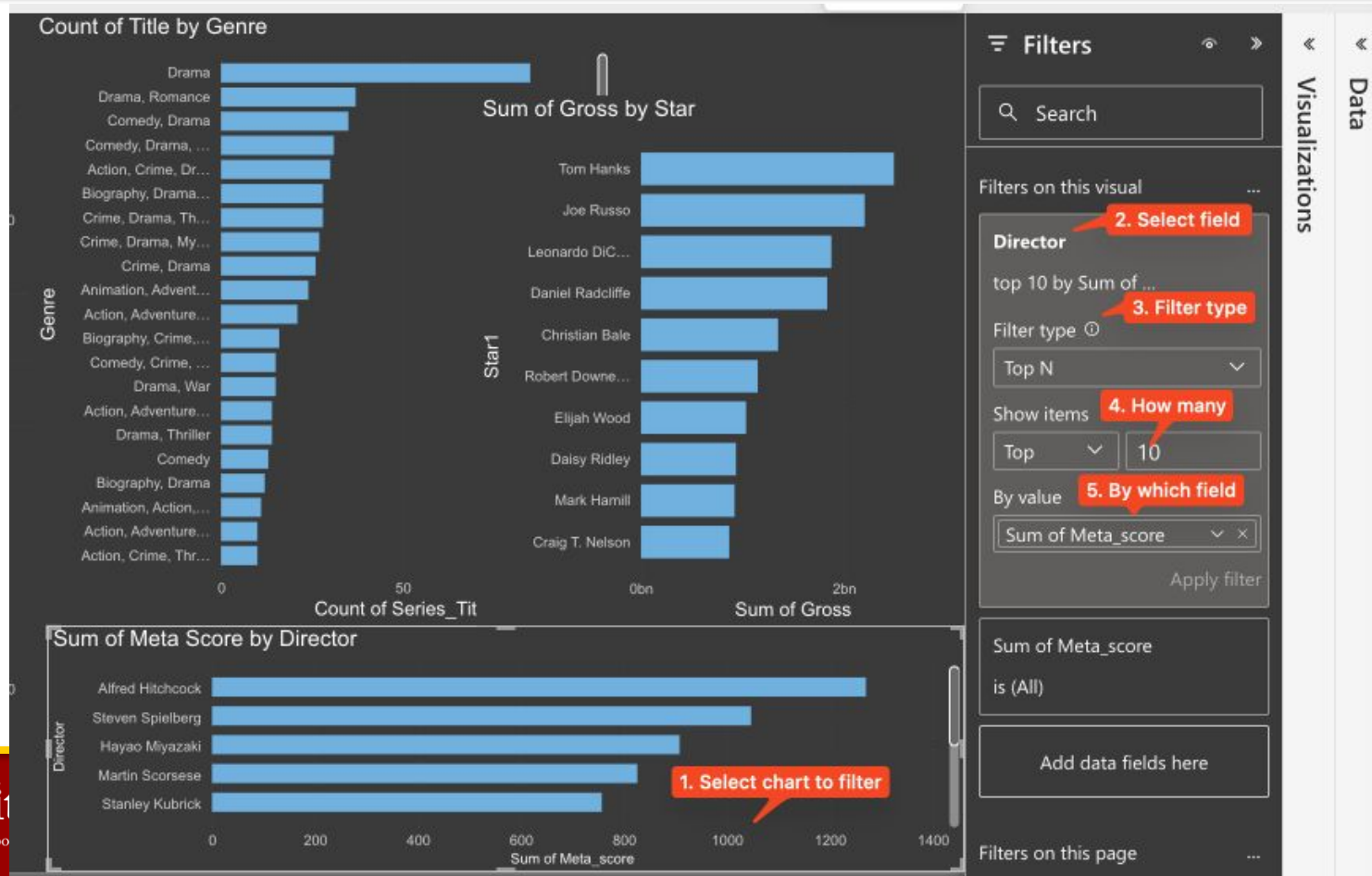
- Dashboards with Interactive Filters: aka (Data) Slicers
- Allow for reporting of raw data metrics, KPIs (exploratory)
- Let users filter and select data themselves w/ user interactive filters / slicers (choose your own / BYO explanatory analysis/viz)

Examples

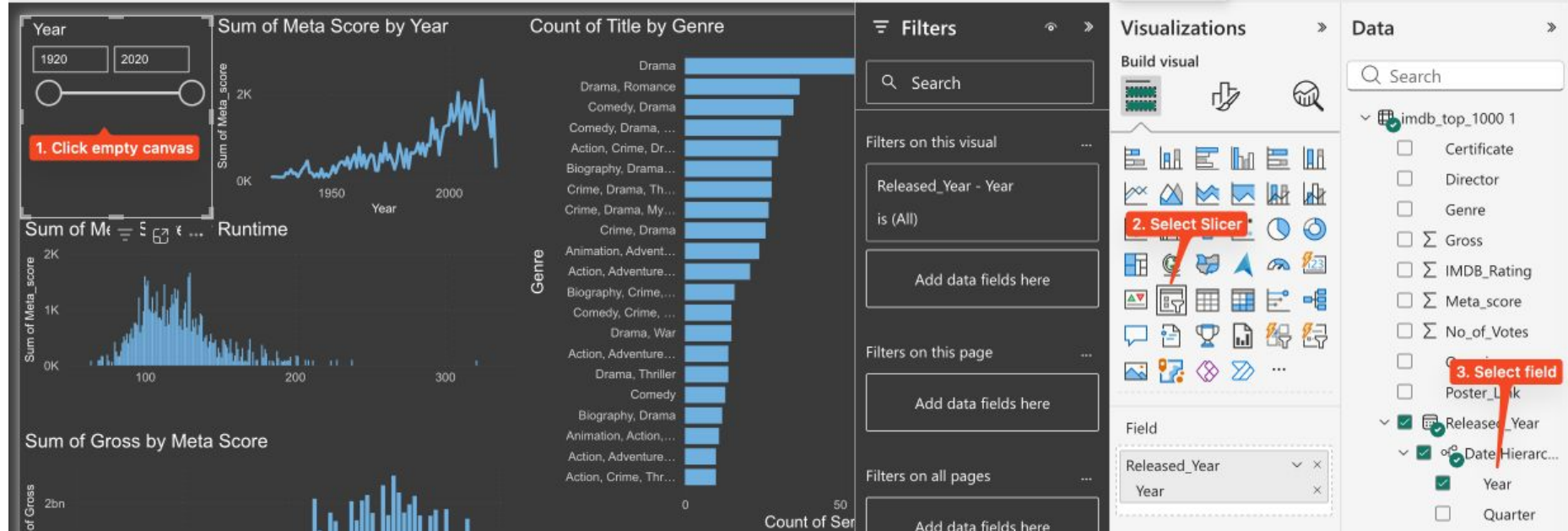


- Trying to find the cause of new / leaving user of a website / app with scatter plots, heat maps, pair plots instead of presenting it w/ highlighting and numbered callouts
- Unsupervised cluster analysis trying to find customer types / groups vs highlighting which group makes the most purchases
- Dashboards with raw data metrics of KPIs vs pre-selected data slicers highlighting / calling out event correlations in time series data or Top N filtering

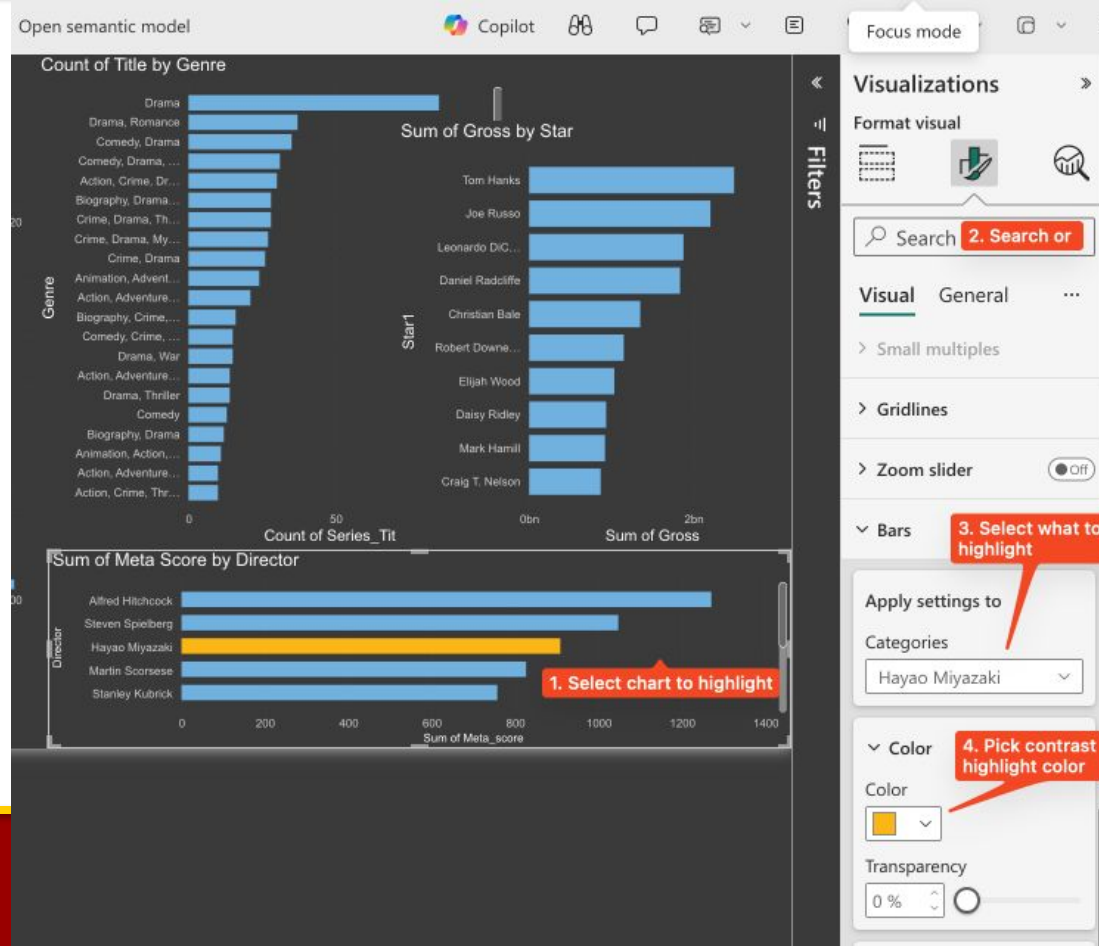
Explanatory Visualization: Filtering in PBI



Explanatory Visualization: Slicing in PBI



Explanatory Visualization: Highlighting in PBI



Google Looker Studio: Start



Looker Studio

Search in Looker Studio

Create

Recent

Shared with me

Owned by me

Trash

Templates

Recent

ReportsData sourcesExplorer

Start with a Template

1. Start here

Blank Report
Looker Studio

GA4 Report
Google Analytics

Acme Marketing
Google Analytics

Search Console Report
Search Console

Template Gallery

Name	Owned by anyone	Last opened by me	Location
Untitled Report	Greg Pohlner	Jan 29, 2026	Owned by me
Untitled Report	Greg Pohlner	Jan 29, 2026	Owned by me
Netflix_Looker_Studio_Report	Greg Pohlner	Jan 28, 2026	Owned by me

Google Looker Studio: Connect Data



Untitled Report
File View Page Help

Reset Share View ? ?

↶ ↷ ↵ 🔍 + Add page + Add data Blend + Add a chart + Add a control <> 🖨️ 📄 🗑️ Theme and layout ⏸ Pause updates

Add data to report

Connect to data My data sources

🔍 Search

Looker
By Google
Connect to your Looker semantic models.

Google Analytics
By Google
Connect to Google Analytics.

Google Ads
By Google
Connect to Google Ads performance report data.

Google Sheets
By Google
Connect to Google Sheets.

BigQuery
By Google
Connect to BigQuery tables and custom queries.

AppSheet
By Google
Connect to AppSheet app data.

CSV File Upload
By Google
Connect to CSV (comma-separated values) files.

Microsoft Excel
By Google
Connect to Microsoft Excel files.

Amazon Redshift
By Google
Connect to Amazon Redshift.

1. We'll be using CSVs and XLSs this week but many connectors are available

Google Looker Studio: Add File



Untitled Report
File View Page Help

Reset Share View ? ?

↶ ↷ ↵ 🔍 + Add page + Add data Blend + Add a chart + Add a control < > Theme and layout Pause updates

← Add data to report Data credentials: Greg Pohlner X

2 GB space available out of 2 GB. [Learn More](#) [Dismiss](#)

CSV CSV File Upload
By Google
You can bring data into Looker Studio by uploading CSV (comma-separated values) files from your computer. Multiple CSV files can be combined as long as the column headers match.
[Learn more](#) [Report an issue](#)

1. Add file

Datasets	Files	Schema
+ Create dataset 🔍 Search datasets Create a dataset by clicking + Create dataset.	+ Add file 🔍 Search files Add files to a dataset by dropping CSV files here or clicking + Add file	

Cancel Add

Google Looker Studio: Auto Schema Ready



Untitled Report
File View Page Help

Reset Share View ? ?

← → ↶ ↷ 🔍 Add page Add data Blend Add a chart Add a control < > Theme and layout Pause updates

← Add data to report Data credentials: Greg Pohlner X

1.98 GB space available out of 2 GB. [Learn More](#) [Dismiss](#)

CSV CSV File Upload
By Google
You can bring data into Looker Studio by uploading CSV (comma-separated values) files from your computer. Multiple CSV files can be combined as long as the column headers match.
[Learn more](#) [Report an issue](#)

Datasets [+ Create dataset](#)

🔍 Search datasets

NetflixDatasetSlides

Files [+ Add file](#)

🔍 Search files

Jan 30, 2026, 4:32 PM	1 file(s) (1 MB/100 MB)
netflix1	1 MB

Schema ⓘ

The fields and their order in this dataset.

- show_id
- type
- title
- director
- country
- date_added
- release_year

1. After processing, the auto schema populates below

2. Click Add

Cancel Add

Google Looker Studio: Add to Report



You are about to add data to this Report

- NetflixDatasetSlides

Note that **Report Editors** can create charts using the new data source(s), and can add dimensions and metrics not currently included in the Report.

Users with access to the report can use its data to generate Google Slides that will be saved in Google Workspace

☐ Don't show me this again

CANCEL

ADD TO REPORT

Google Looker Studio: Blank Canvas



Untitled Report

File Edit View Insert Page Arrange Resource Help

Reset Share View ? ?

Undo Redo Select Search Add page Add data Blend Add a chart Add a control Add a filter Theme and layout Pause updates

Add filter

Freeform layout Responsive layout

Let's get started

Drag a field from the Data Panel to the canvas to add a new chart or select a component on the report canvas to edit it.

Data

Search

NetflixDatasetSlides

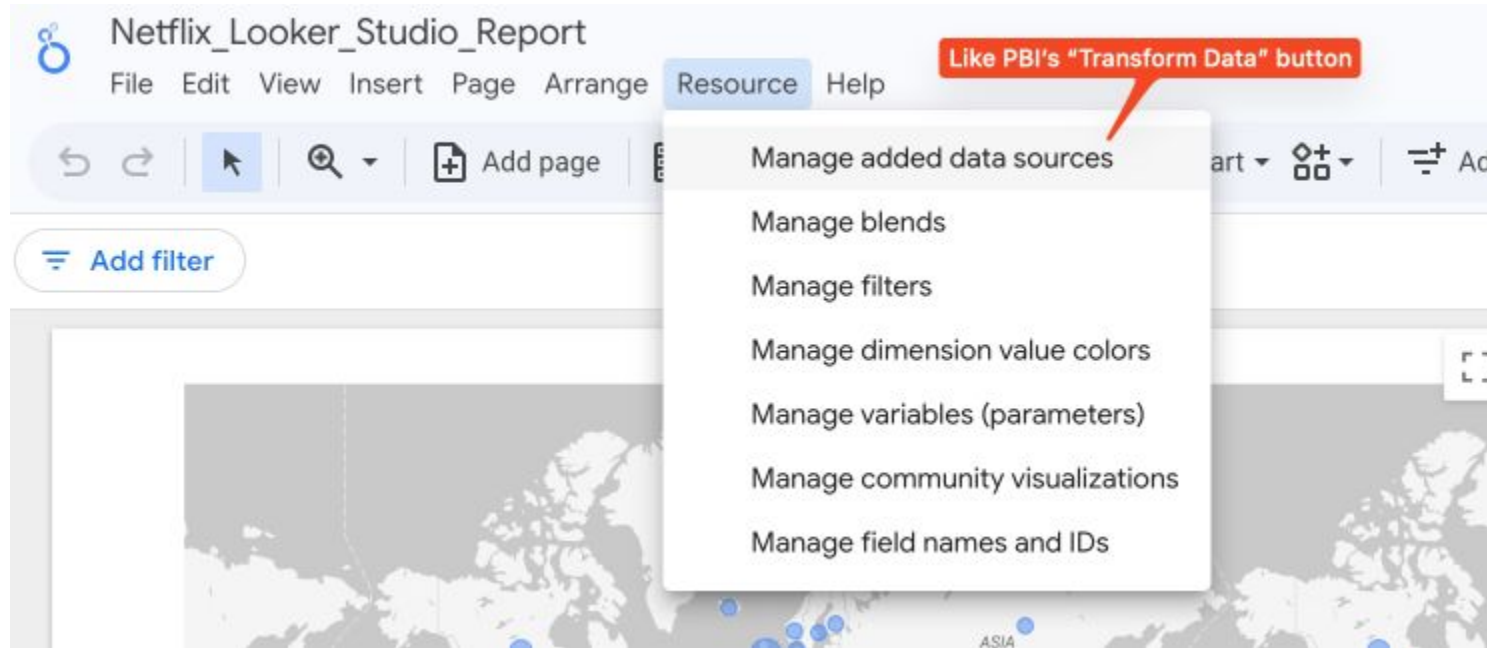
- country
- date_added
- director
- duration
- listed_in
- rating
- release_year
- show_id
- title
- type
- Record Count

+ Add a field

+ Add a parameter

+ Add Data

Google Looker Studio: Transform Data



Google Looker Studio: Edit Data



Netflix_Looker_Studio_Report

File Edit View Insert Page Arrange Resource Help

Reset Share View ? [Profile]

Data sources [X] CLOSE

Name	Connector Type	Type	Used in report	Status	Actions	Alias
Netflix Dataset 2	CSV File Upload	Embedded	2 charts, 0 variables	Working	EDIT DUPLICATE REMOVE MAKE REUSABLE	ds0

+ ADD A DATA SOURCE

Then Edit

Google Looker Studio: Model View



Netflix_Looker_Studio_Report

File Edit View Insert Page Arrange Resource Help

Netflix Dataset 2 **Similar to PBI Semantic Model View** Scope: Embedded Data credentials: Viewer Data freshness: ...

Edit Connection Add a field Add a parameter Filter by email

*This data source is embedded in the report, so **report viewers** have access to data source metadata, including column names, the connection settings, and the data source type. To restrict access to connection settings, make this data source reusable.*

Field	Type	Default Aggregation	Description
Dimensions (13)			
country	Country	None	
date_added	Text	None	
date_added_as_date	Date	None	
date_added_as_date_2	Date	None	
director	Text	None	
duration	Text	None	
listed_in	Text	None	
rating	Text	None	
release_year	Number	Sum	
release_year_as_date	Date	None	
show_id	Text	None	
title	Text	None	
type	Text	None	
Metrics (2)			
Record Count	Number	Auto	

REFRESH FIELDS All Fields Data Preview

Google Looker Studio: Add Calc Field



Netflix_Looker_Studio_Report

File Edit View Insert Page Arrange Resource Help

← Netflix Dataset 2 Scope: [icon]

← Edit Connection **+ Add a field** + Add a parameter Filter by email

1. Click Add a field

(i) This data source is embedded in the report, so **report viewers** have access to data source. To restrict access to connection settings, make this data source reusable.

Field ↑	Type
Dimensions (13)	
country	Country
date_added	Text
date_added_as_date	Date

Netflix_Looker_Studio_Report

File Edit View Insert Page Arrange Resource Help

← Netflix Dataset 2

← Edit Connection **+ Add a field** + Add a parameter Filter by email

2. Add calc field

Add calculated field

Add group

Add bin

(i) This data source is embedded in the report, so **report viewers** have access to data source. To restrict access to connection settings, make this data source reusable.

Field ↑	Type
Dimensions (13)	
country	Country
date_added	Text

Google Looker Studio: Parse Date



 Netflix_Looker_Studio_Report

File Edit View Insert Page Arrange Resource Help

← Netflix Dataset 2

Scope: Embedded Data credentials

← All Fields

Available Fields

Search

country

ABC date_added

date_added_as_date

date_added_as_date_2

ABC director

ABC duration

ABC listed_in

ABC rating

123 release_year

release_year_as_date

ABC show_id

ABC title

ABC type

123 Record Count

123 title_count

Field Name

e.g. New Calculated Field

date_added_as_date_2

Formula

1 PARSE_DATE('%m/%d/%Y', date_added)

SAS Viya: Start



[Contact Us](#)  [Software](#) [Learn](#) [Support](#) [Partners](#) [About Us](#) [SAS Innovate](#)

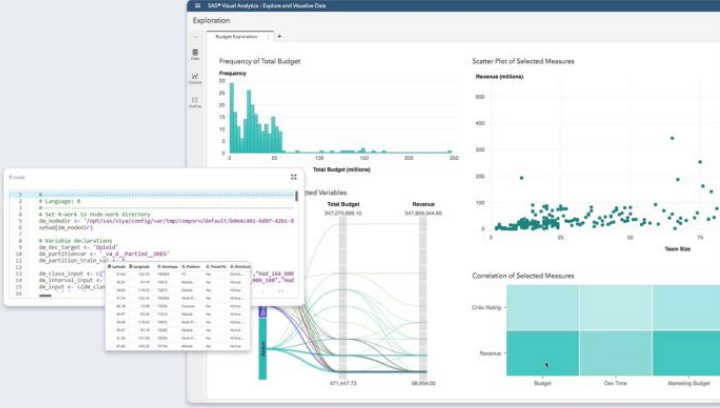
    

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School of https://www.sas.com/en_us/contact_geo.html

california

SAS Viya: Acknowledge



All SAS ▾



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Overview Programs ▾ Corporate Training My Training



My Training

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- SAS Viya for Learners

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1

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2

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What is...

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SAS Viya: (Re-)Start



The first login may take a while while it creates your student cloud account, and then eventually bring you to a course catalog when it finally completes.

Try this direct link: <https://learn.sas.com/course/view.php?id=6825>



 | LEARN

All SAS 

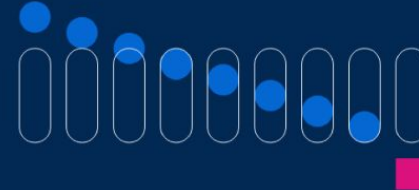
Courses  What Do You Want to Learn? 

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Gregory Pohlner  

SAS® Viya® for Learners

Build a complete advanced analytics program - from entry-level to doctorate - with a single software environment.



SAS Viya for Learners 4

1

Launch Software

Access JupyterLab

Helpful Tips

- Use Chrome, Firefox, or Safari to access SAS Viya for Learners.
- SAS Viya for Learners is a shared resource.
 - Inactive accounts and files may be deleted to ensure maximum performance.
 - Users are limited to 5 GB of data in memory and 100 MB per uploaded data file.
- Autotuning is disabled in SAS Viya for Learners. The software will return an error when a user requests autotuning.
- SAS Viya for Learners may be unavailable every 3rd weekend of the month from

Additional Information

- Get up and running quickly with [SAS Viya Quickstart Videos](#).
- Find documentation on existing shared data sets in the SAS Viya for Learners [Data Repository](#).



SAS Viya: Landing



 ? P

 Show all applications

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Data & AI Life Cycle Quick Start



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Discover Information Assets



SAS® STUDIO
Develop Code and Flows



SAS® VISUAL ANALYTICS
Explore and Visualize



MODEL STUDIO
Build Models



SAS® MODEL MANAGER
Manage Models

SAS Viya: Explore and Visualize



SAS® Visual Analytics - Explore and Visualize

Explore and Visualize

Explore data, apply predictive analytics, and build interactive reports.

New report

1. Navigate through this folder path

Search

SAS Content > Courses > VISUAL > practices

	Name	Date Created	Created By	Date Modified
☐	120_720_Creating a...	02/27/24 12:46 ...	designer	06/01/24 5:15
☐	130_104_Using Latti...	11/17/23 10:11 ...	designer	06/01/24 5:15
☐	130_107_Modifying...	11/28/23 2:44 PM	designer	06/01/24 5:15
☐	130_203_Creating L...	02/27/24 12:46 ...	designer	06/01/24 5:15
☐	130_303_Using Hist...	02/28/24 11:43 ...	designer	06/01/24 5:15
☐	130_404_Working w...	03/05/24 9:56 AM	designer	06/01/24 5:15
☐	130_503_Creating L...	03/07/24 2:53 PM	designer	06/01/24 5:15
☐	140_102_Saving an ...	11/29/23 4:59 PM	designer	06/01/24 5:15
☐	140_203_Creating T...	12/01/23 8:02 AM	designer	06/01/24 5:15
☐	140_303_Creating a...	12/06/23 8:26 AM	designer	06/01/24 5:15
☐	140_404_Creating a...	12/14/23 8:59 AM	designer	06/01/24 5:15
☐	140_503_Working w...	12/19/23 9:17 AM	designer	06/01/24 5:15
☐	140_603_Working w...	12/20/23 2:14 PM	designer	06/01/24 5:15
☐	140_703_Using Pro...	01/23/24 2:01 PM	designer	06/01/24 5:15

2. Scroll down and Double click this one

Opened Reports

- 101_142_Using Calculated It...
- 102_111_Viewing a Report ...
- 140_203_Creating Targeted ...

What's New

Read about the latest updates

Learn More

Beautiful Reports

SAS Viya: Blank View Mode



SAS® Visual Analytics - Explore and Visualize

140_303_Creating a Dual Axis Bar-Line Chart

Open reports (4)

Page 1 Page 2 Page 3 Page 4

click the edit (pen) button in upper left

No content to display.

SAS Viya: Add Data



SAS® Visual Analytics - Explore and Visualize

Editing 140_303_Creating a Dual Axis Bar-Line Chart

Opened reports (4)

Data

INSIGHT_TOY_COMPANY_2017

Filter

- + New data item
- Category
 - Facility Age - 2
 - Facility Opening Date - 19
 - Transaction Month - 12
- Measure
 - Facility Efficiency
 - Facility Employees
 - Frequency
- Aggregated Measure
 - Frequency Percent

Add data

- Import data
- New data from join
- New data from join with INSIGHT_TOY_COMPANY_2017
- New data from aggregation of INSIGHT_TOY_COMPA...
- Save data view...
- Manage data views
- Remove INSIGHT_TOY_COMPANY_2017
- Replace INSIGHT_TOY_COMPANY_2017
- Refresh INSIGHT_TOY_COMPANY_2017
- View INSIGHT_TOY_COMPANY_2017 source table
- Apply data filter
- Map data
- Set unique row identifier

Design a Report

objects or data items onto the page, or start from a page template.

Select a template

SAS Viya: Import Data



SAS® Visual Analytics - Explore and Visualize

Editing 140_303_Creating a Dual Axis Bar-Line Chart

Dual Axis Bar-Line Chart Page 2 Page 3 Page 4 +

Opened reports (4)

Choose Data

Search all data

Show: Available (402)

☐	Name	★	Library	Date Modified	Modif
☐	AAEM51_TRANSACTIONS	☆	ACADEMIC	Jan 16, 2026 3:50 PM	sas
☐	AATESTFILE001	☆	ACADEMIC	Jan 16, 2026 3:50 PM	sas
☐	ABSENTEEISM	☆	ACADEMIC	Jan 16, 2026 3:50 PM	sas
☐	ACACIA	☆	ACADEMIC	Jan 16, 2026 3:49 PM	sas
☐	ACCEPTS	☆	ACADEMIC	Jan 16, 2026 3:50 PM	sas
☐	ACCIDENTAL_DRUG_DEAT...	☆	ACADEMIC	Jan 16, 2026 3:50 PM	sas
☐	ACCUMDATA	☆	ACADEMIC	Jan 16, 2026 3:50 PM	sas

402 of 402

AAEM51_TRANSACTIONS

Details Columns (4)

Properties

Asset type: In-memory data

Date modified: Jan 16, 2026 3:50 PM

Modified by: sas

Date created: Jan 16, 2026 3:50 PM

Created by: sas

Source table: aaem51_transactio...

Add Cancel

1 Import data

SAS Viya: Local Files



SAS® Visual Analytics - Explore and Visualize

Editing 140_303_Creating a Dual Axis Bar-Line Chart

Opened reports (4)

Import Data

Imports + Add files

Select local files or files from SAS content to import them.

1 Local files SAS content

Drag local files here to import them.

Add to report Cancel

SAS Viya: Target Location



SAS® Visual Analytics - Explore and Visualize

Editing 140_303_Creating a Dual Axis Bar-Line Chart

Opened reports (4)

Import Data

Imports + Add files

netflix1.csv
Local file

netflix1.csv

Target table name: *
netflix1

Target location: *

If target table name exists:
☒ Cancel import
☐ Replace file

Input file delimiter:
Comma

Scanned rows: * ⓘ
20

Locale:
Enter locale

Import item Import all

Add to report Cancel

1

SAS Viya: Target Location User folder



The screenshot displays the SAS Viya interface with the 'Import Data' dialog box open. The 'Imports' section shows a file named 'netflix1.csv' as a 'Local file'. A 'Choose a Library' sub-dialog is also open, showing a list of libraries. The 'CASUSER(pohlner@usc.edu)' library is selected, highlighted by a red circle with the number 2. The 'OK' button is highlighted by a red circle with the number 3. The background shows the 'Editing' tab for '140_303_Creating a Dual Axis Bar-Line Chart'.

Import Data

Imports

- netflix1.csv
Local file

Choose a Library

- cas-shared-default
 - ACADEMIC
 - ADML
 - CASUSER(pohlner@usc.edu)**
 - CORTEXfr
 - CPML
 - CRVA83
 - Formats
 - FVVF
 - GAINDS
 - GAVA85
 - HELPPDATA
 - INTN
 - ModelPerformanceData
 - Models
 - ORVY
 - PIVY

Buttons: Import item, Import all, OK, Cancel, Add to report, Cancel

SAS Viya: Import All



SAS® Visual Analytics - Explore and Visualize

Editing 140_303_Creating a Dual Axis Bar-Line Chart

Opened reports (4)

Import Data

Imports + Add files

netflix1.csv
Local file

netflix1.csv

Target table name: *
netflix1

Target location: *
cas-shared-default/CASUSER(pohlner@usc.e...

If target table name exists:
☒ Cancel import
☐ Replace file

Input file delimiter:
Comma

Scanned rows: * ⓘ
20

Locale:
Enter locale

Import item Import all

Add to report Cancel

SAS Viya: Add To Report



SAS® Visual Analytics - Explore and Visualize

Editing 140_303_Creating a Dual Axis Bar-Line Chart

Opened reports (4)

Import Data

Imports + Add files

netflix1.csv
Local file

netflix1.csv **Import item** **Import all**

✓ The table was successfully imported on 1/30/2026 6:02:00 PM and is ready to use.

Target table name: *

netflix2

Target location: *

cas-shared-default/CASUSER(pohlner@usc.e...

If target table name exists:

☒ Cancel import

☐ Replace file

Input file delimiter:

Comma

Scanned rows: * @

20

1 **Add to report** **Cancel**

SAS Viya: Blank Canvas



SAS® Visual Analytics - Explore and Visualize

Editing 140_303_Creating a Dual Axis Bar-Line Chart

Opened reports (4)

Dual Axis Bar-Line Chart Page 2 Page 3 Page 4 : +

Data

NETFLIX2

Filter

+ New data item

Category

- country - 86
- date_added - 1.7K
- director - 4.5K
- duration - 220
- listed_in - 513
- rating - 14
- show_id - 8.8K
- title - 8.8K
- type - 2

Measure

- Frequency
- release_year

Aggregated Measure

- Frequency Percent

Design a Report

Drag objects or data items onto the page, or start from a page template.

Select a template

SAS Viya: Parse Date



SAS® Visual Analytics - Explore and Visualize

Editing 101_122_Using Suggestions Pane

Data

NETFLIX1_2

1. New data item -> New Calculated Item

+ New data item

Category

Calculated

country - 8

date_adde

date_adde

director - 4

duration -

listed_in -

rating - 14

show_id - 8

title - 8.8K

type - 2

Geography

Country_a

Measure

Frequency

release_ye

New Calculated Item

Name: * 2. Name it

3. Functions -> Search "parse"

date_added_as_date

DATE9

Operators Functions Data New parameter

1 Parse(date_added, 'MMDDYY32')

Results

date_added	date_added_as_date
9/25/2021	September 25, 2021
9/24/2021	September 24, 2021

Preview selection only

Parse

4. Select input column

value:

date_added

format:

5. Select "MMDDYY32" format

Parse

interprets a numeric or datetime value from the input string. **format** specifies the format that is used to interpret the string. The output from the Parse function is either a number or a datetime value,

OK Cancel

Pro Code Visualization over No Code vs Low Code



- **Applicability**
 - Vendor lock-in
 - Tool-specific
- **Expressivity**
 - String methods read like english, method chaining
- **Control**
 - Deep Customization; Convention vs Configuration
- **Flexibility**
 - Multiple ways to do things in general purpose languages
- **Ecosystem / Community**
 - Libraries, packages, extensions, plug-ins
- **Scalability**
 - Same code on multiple instances of multiple datasets; Write once, run many times
- **Extensibility**
 - Easy to add to existing code
- **Maintainability**
 - Tooling, visibility
- **Ease of Automation**
 - No mouse, no point and click, HITL optional

In class demos



- More PBI
- Comparison to
 - SAS Viya
 - Google Looker Studio
 - Tableau



Resources and References

- Power BI Login: <https://app.powerbi.com/>
- [MSFT Power BI Tutorial](#)
 - [MSFT Financial Sample Data Set](#)
 - [More MSFT Sample Data](#)
- [Kaggle IMDB Data Set](#)
- [Kaggle Netflix Data Set](#)

Questions?

